

Abdullah Ejaz Janjua

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EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

Bachelor of Science in Computer Science (CGPA: 3.78/4.00)

Topi, Pakistan

Sep. 2023 – June 2027

EXPERIENCE

Mitacs Global Research Intern

Carleton University

June 2026 – Present

Ottawa, Canada

- Awarded a fully funded Mitacs Globalink Research Internship to conduct research under the supervision of Dr. Junfeng Wen.
- Adapting pre-trained diffusion models along the temporal axis to model continuous facial aging while preserving identity consistency.
- Conducting comparative experiments against existing generative baselines to assess trade-offs between synthesis realism and temporal identity retention.

Teaching Assistant

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

January 2026 – May 2026

Topi, Pakistan

- Assisting 40+ students in the Verilog Lab under Dr. Taj Khan by troubleshooting digital logic and HDL programming errors.
- Led a 3-day Linux Lab Tutorial focused on terminal operations, shell navigation, and command-line workflows.
- Identified areas where students required more time and proposed schedule modifications to the professor to ensure better comprehension.
- Conducted technical vivas to verify students' understanding.

Undergraduate Researcher

Intelligent Machines Lab, Information Technology University

June 2025 - November 2025

Lahore, Pakistan

- Worked under the guidance of Dr. Waqas Sultani on Domain Generalization, specifically investigating methods to improve DETR-based models' generalizability to unseen target domains
- Improved generalization of the model from 34.1 to 38.7 mAP@50 on the LeukemiaAttri dataset by proposing incorporating Discrete Fourier Transform (DFT) with model features.
- Converted my findings into a comprehensive academic paper.

Research Intern

Intelligent Machines Lab, Information Technology University

Jan 2025 – Feb 2025

Lahore, Pakistan

- Worked with Dr. Waqas Sultani's research group, where I helped develop an Unsupervised Domain Adaptation (UDA) framework for microscopy images with source-target domain transfer.
- Proposed and helped with integration of Monte Carlo Dropout (MCD) for uncertainty-aware pseudo-labeling and confidence image-mixing.
- Assisted in writing, running experiments and designing experiments.

Artificial Intelligence Intern

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

Sep 2024 – Jan 2025

Topi, Pakistan

- Worked under the guidance of Dr. Khurram Khan Jadoon, where I developed and implemented object detection and segmentation algorithms for weapon identification.
- Designed and executed a detection algorithm for Foreign Object Detection (FoD A).
- Created and manually annotated a comprehensive weapon dataset for training high-performance segmentation models.
- Achieved competitive performance compared to industry benchmarks in security applications.

PROJECTS

TinyTorch: Custom C++/CUDA Deep Learning Framework

- Developed a C++ deep learning framework from scratch with Python bindings (pybind11), featuring a custom DAG-based autograd engine utilizing topological sort for backpropagation.
- Engineered custom CUDA kernels for 2D Convolutions (forward and transposed backward passes), Linear layers, and GAP, utilizing shared memory tiling and 1D stride loops to manage memory coalescing.
- Built a specialized, fused Softmax-Cross Entropy kernel that executed faster than PyTorch's generalized equivalent
- Trained a custom Conv-ReLU-Conv-Linear architecture on the MNIST dataset using a custom SGD optimizer and manual GPU memory management, achieving 95% test accuracy in 20 minutes.
- **GitHub:** [TinyTorch](#)

Generative AI: VAE & RealNVP

- Implemented a Variational Autoencoder (VAE) to study latent variable models and probabilistic image generation
- Built a RealNVP flow-based model utilizing affine coupling layers for exact density estimation and efficient sampling.
- Analyzed manifold learning and generative capabilities through latent space visualization and reconstruction metrics.
- **GitHub:** [VAE](#), [RealNVP](#)

Self-Supervised Learning (DINOv1)

- Implemented the DINOv1 training pipeline for ResNet models, enabling self-supervised feature learning without labels.
- Developed a finetuning pipeline to adapt the pre-trained model for classification tasks.
- Utilized clear coding practices, allowing for seamless training of the model from the CLI without modifying code.
- **GitHub:** [DINOv1-ResNet](#)

Foundational CV Architectures: ViT & ResNet

- Implemented Vision Transformer (ViT) and ResNet (50, 101, 152) architectures from scratch in PyTorch, adhering to original research specifications.
- Engineered core components including multi-head self-attention, positional encoding, patch embeddings, and residual skip connections.
- Validated models on PascalVOC2012 and CardDataset, achieving 70.66% and 88.44% accuracy respectively without pre-training or hyperparameter tuning.
- **GitHub:** [ViT](#), [ResNet](#)

SKILLS

Languages: Python, C, C++, CUDA-C, SQL, HTML, CSS

Soft Skills: Academic Writing, Communication, Presentation

Libraries: NumPy, Pandas, Matplotlib, OpenCV, scikit-learn, PyTorch

COURSEWORK

University Courses: Object-Oriented Programming, Data Structures & Algorithms, GPU Programming, Operating Systems, Linear Algebra, Linear Algebra II, Statistics, Calculus I

Self-Study: CS50x: Introduction to Computer Science (Prof. David Malan), Machine Learning Specialization (Prof. Andrew Ng), CS231n: Deep Learning for Visual Recognition (Prof. Fei-Fei Li), Computer Vision (Prof. Andreas Geiger, Diffusion Models (Prof. Stefano Ermon))

ACHIEVEMENTS

Dean's Honor Roll: Achieved High Distinction for all semesters so far in recognition of outstanding academic performance. (SGPA ≥ 3.75)

Awarded shield for highest marks obtained in HSSC in my college.